

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I _K Sreelatha(192425248)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sree.K

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s=255-r$ to the pixel values:

$r=[50,100,150,200]$

Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s=c\log_{10}(1+r)$, where $c=10$, compute s for:

$r=[0,10,50,100]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

(b) Given a 3-bit image with histogram:

Intensity Frequency

0 2

1 2

2 4

3 8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement

Grey Level Transformation is a **point processing technique** in digital image processing where each pixel value is modified independently based on a mathematical function. It changes the brightness or contrast of an image without considering neighboring pixels.

Purpose

- Enhances image quality for better visualization.
- Improves brightness and contrast.
- Highlights important image features.
- Makes dark images brighter or bright images darker.
- Prepares images for further image processing tasks such as segmentation, object detection, and feature extraction.

Applications

- Medical imaging (X-ray, MRI, CT scan)
- Satellite image enhancement
- Industrial inspection
- Document image processing
- Security and surveillance systems

Types of Grey Level Transformations

1. Image Negative
2. Log Transformation
3. Power-Law (Gamma) Transformation
4. Piecewise Linear Transformation
5. Contrast Stretching
6. Thresholding

(b) Apply Negative Transformation

Formula:

$$s = 255 - r$$

Where:

- **r** = Original pixel value
- **s** = Transformed pixel value

Given:

$$r = [50, 100, 150, 200]$$

Calculation

For r = 50

$$s = 255 - 50 = 205$$

For r = 100

$$s = 255 - 100 = 155$$

For r = 150

$$s = 255 - 150 = 105$$

For r = 200

$$s = 255 - 200 = 55$$

Result

Original (r) Negative (s)

50 205

100 155

150 105

200 55

Answer: [205, 155, 105, 55]

Advantages

- Highlights white details on dark backgrounds.
- Useful in medical images.
- Makes hidden details visible.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels

Log Transformation is used to expand dark pixel values while compressing bright pixel values.

Formula:

$$s = c \log(1 + r)$$

where

- **r** = Input pixel intensity
- **c** = Scaling constant
- **s** = Output pixel intensity

Purpose

- Enhances details in dark regions.
- Compresses high intensity values.
- Improves visibility of low brightness areas.
- Widely used in Fourier spectrum display.

Characteristics

- Small intensity values increase significantly.
- Large intensity values increase slowly.
- Produces brighter images.

Applications

- Medical imaging

- Astronomy
- Satellite images
- Fingerprint enhancement

(b) Compute Log Transformation

Given

$$c = 10$$

Use natural logarithm (ln).

For r=0

$$s = 10\ln(1) = 10(0) = 0$$

For r=10

$$\begin{aligned} s &= 10\ln(11) \\ &= 10(2.3979) = 23.98 \end{aligned}$$

For r=50

$$\begin{aligned} s &= 10\ln(51) \\ &= 10(3.9318) = 39.32 \end{aligned}$$

For r=100

$$\begin{aligned} s &= 10\ln(101) \\ &= 10(4.6151) = 46.15 \end{aligned}$$

Result

r s

0 0

10 23.98

50 39.32

r s

100 46.15

Final Answer

$[0, 23.98, 39.32, 46.15]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement

Histogram Equalization is a contrast enhancement technique that redistributes pixel intensity values uniformly over the available range.

Purpose

- Improves image contrast.
- Makes hidden details visible.
- Enhances low contrast images.
- Spreads pixel intensities across the entire range.

Working Steps

1. Compute histogram.
2. Compute cumulative histogram (CDF).
3. Normalize the CDF.
4. Map old intensities to new intensities.

Applications

- Medical images
- Remote sensing
- Face recognition
- Image preprocessing
- Object detection

Advantages

- Simple and effective.
- Automatically enhances contrast.
- Improves visualization.

(b) Compute Histogram Equalization

Given Histogram

Intensity Frequency

0	2
1	2
2	4
3	8

Total Pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Maximum gray level

$$L = 4$$

Step 1: Cumulative Frequency

Intensity Frequency CDF

0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Intensity

Formula

$$s = \frac{CDF}{N}(L - 1)$$

Since

$$L - 1 = 3$$

For intensity 0

$$\frac{2}{16} \times 3 = 0.375 \approx 0$$

For intensity 1

$$\frac{4}{16} \times 3 = 0.75 \approx 1$$

For intensity 2

$$\frac{8}{16} \times 3 = 1.5 \approx 2$$

For intensity 3

$$\frac{16}{16} \times 3 = 3$$

Result

Old Intensity CDF Equalized Intensity

0	2	0
1	4	1
2	8	2
3	16	3

Interpretation

Since the histogram is already fairly distributed, histogram equalization produces only minor changes.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing

Spatial filtering performs operations directly on image pixels using a small mask (kernel).

A smoothing filter reduces intensity variations between neighboring pixels.

Purpose

- Removes noise.
- Reduces random variations.
- Produces a blurred image.
- Smooths object boundaries.
- Improves image quality before segmentation.

Common Filters

- Mean (Average) Filter
- Gaussian Filter
- Median Filter

Applications

- Noise removal
- Medical imaging
- Face detection
- Image preprocessing

(b) Apply 3×3 Averaging Filter

Matrix

10	20	30
20	30	40
30	40	50

Average Filter

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Sum of all pixels

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average

$$\frac{270}{9} = 30$$

New Center Pixel

$$\boxed{30}$$

Interpretation

The averaging filter smooths the image by replacing the center pixel with the average of its neighboring pixels, reducing noise but also slightly blurring edges.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details

Sharpening increases the visibility of edges and fine details by emphasizing rapid changes in intensity.

The Laplacian operator is a **second-order derivative** filter that highlights edges in all directions.

Purpose

- Enhances edges.
- Increases image clarity.
- Improves object boundaries.
- Highlights fine details.
- Makes blurred images appear sharper.

Common Sharpening Filters

- Laplacian Filter
- High-pass Filter
- Unsharp Masking
- High-boost Filtering

Applications

- Medical diagnosis
- Satellite imaging
- Fingerprint recognition
- Industrial quality inspection

(b) Apply Laplacian Mask

Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Calculation

Center pixel:

$$4 \times 20 = 80$$

Top:

$$-1 \times 10 = -10$$

Bottom:

$$-1 \times 10 = -10$$

Left:

$$-1 \times 10 = -10$$

Right:

$$-1 \times 10 = -10$$

Corners:

$$0 \times 10 = 0$$

Total

$$80 - 10 - 10 - 10 - 10 = 40$$

Output at Center

$$\boxed{40}$$

Interpretation

A positive output indicates that the center pixel differs significantly from its neighbors, meaning it is part of an edge or fine detail. The Laplacian filter enhances such regions, making edges sharper and more prominent.